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Department of CSE

B.Tech. CSE Trimester -5

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Section -3

Cloud Infrastructure and Services

25CS2206E

SKILL RECORD

2026

Skill -1

AIM: To create an IAM user with programmatic access only, download the access key and secret key, and set up a billing alert for the account so that an email is received when the Free Tier limit is about to be crossed.

Description:

AWS Identity and Access Management (IAM) is a web service that is used to securely control access to AWS resources. It is used to create and manage AWS users and user groups, and to apply permissions that allow or deny their access to AWS resources. An IAM user is an identity that is created within an AWS account. Programmatic access allows a user to call AWS services through the AWS CLI, the AWS SDKs or direct API calls using an access key, without signing in to the AWS Management Console.

An access key consists of two parts, an Access Key ID and a Secret Access Key, which together act like a user name and password for the programmatic calls. The secret access key can be viewed or downloaded only once at the time of creation, so it has to be stored safely. While creating an IAM user, the permissions can be attached directly to the user, copied from an existing user, or the user can be added to a user group. In this skill, the AdministratorAccess AWS managed policy is attached to the user.

Since the account is a free tier account, unexpected charges have to be avoided. AWS Budgets in the Billing and Cost Management console is used to set up a billing alert. A zero spend budget sends an email notification to the given email recipients whenever the spending exceeds $0.01, so that the account owner is alerted before the free tier limit is crossed and any charges are incurred.

Steps:

1. Sign in to the AWS Management Console and open the IAM (Identity and Access Management) service.

2. In the IAM console, go to Access Management and choose IAM users, then click Create user.

3. In the Specify user details step, enter the user name as Shiva and keep the option Provide user access to the AWS Management Console unchecked, so that the user is created with programmatic access only. Then click Next.

4. In the Set permissions step, select Attach policies directly, search for the

AdministratorAccess policy and select it. Then click Next.

5. In the Review and create step, verify the user details such as the user name, console password type as None and the attached AdministratorAccess policy, then click Create user.

6. To test the created user, console access was enabled from the Security credentials tab and the autogenerated console password was downloaded as a CSV file.

7. Sign in to the AWS console as the IAM user using the account ID 741973715162 and the IAM user name on the sign-in page.

8. After signing in, the Console Home page opened for the IAM user, which confirmed that the user was created correctly.

9. Open the IAM user Shiva, go to the Security credentials tab and click Create access key.

10. In the Access key best practices and alternatives step, select the Command Line Interface (CLI) use case and continue.

11. In the Retrieve access keys step, the Access Key ID and the Secret Access Key were displayed. The credentials were downloaded as a CSV file with the Download .csv file button, because the secret access key cannot be viewed again later.

12. The Security credentials page of the user Shiva showed Access keys (1) with the status Active.

13. A second IAM user New was created in the same way, with console access kept disabled, so that the user has programmatic access only.

14. For the user New, click Create access key, select the Command Line Interface (CLI) use case and proceed to retrieve the access keys.

15. The Access Key ID and Secret Access Key of the user New were displayed only once and were downloaded as a CSV file.

16. The Security credentials page of the user New confirmed that the access key was created with the status Active.

17. The IAM users page listed both the users New and Shiva with their details.

18. To set up the billing alert, search for billing in the AWS console search bar and open Billing and Cost Management.

19. In the Billing console, go to Budgets and click Create budget. The Use a template (simplified) option with the Zero spend budget template was reviewed, and the Customize (advanced) option with the Cost budget was also reviewed.

20. Finally, the Zero spend budget template was selected. The budget name was entered as My Zero-Spend Budget and the email recipient was given as karkalashivareddy@gmail.com. With all AWS services in scope, an email notification is sent when any spend above $0.01 is incurred. Then click Create budget to complete the setup.

Result: Thus, an IAM user with programmatic access only was created successfully, the access key and secret access key were generated and downloaded, and a billing alert was set up using a zero spend budget so that an email is received when the spending is about to cross the free tier limit.

Output:

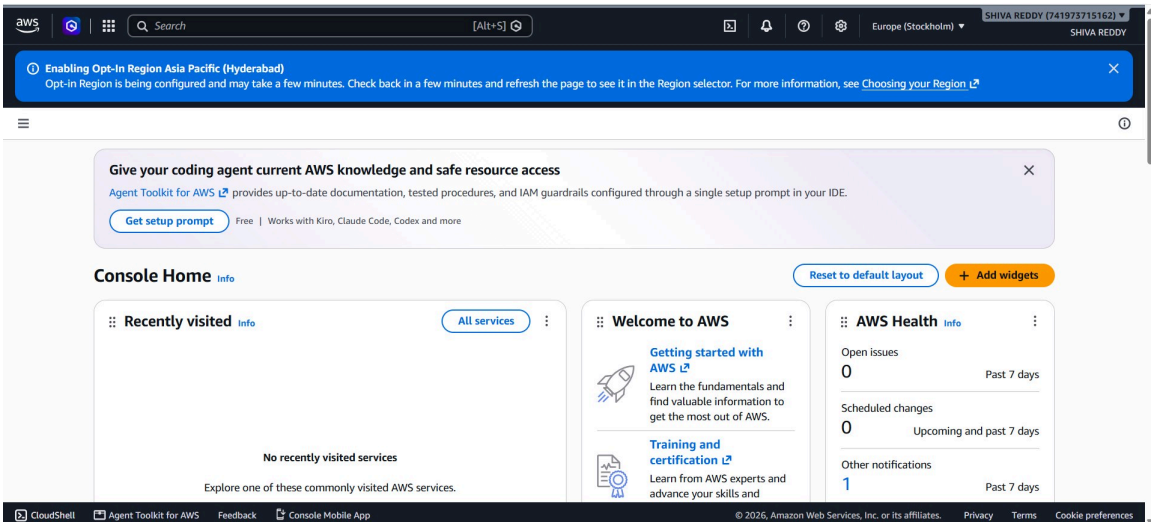


Fig 1: AWS Management Console Home page

The AWS Management Console home page opened after signing in. The account name SHIVA REDDY with the account ID 741973715162 is shown at the top right corner and the region selected was Europe (Stockholm). From here the IAM service was opened to create the user.

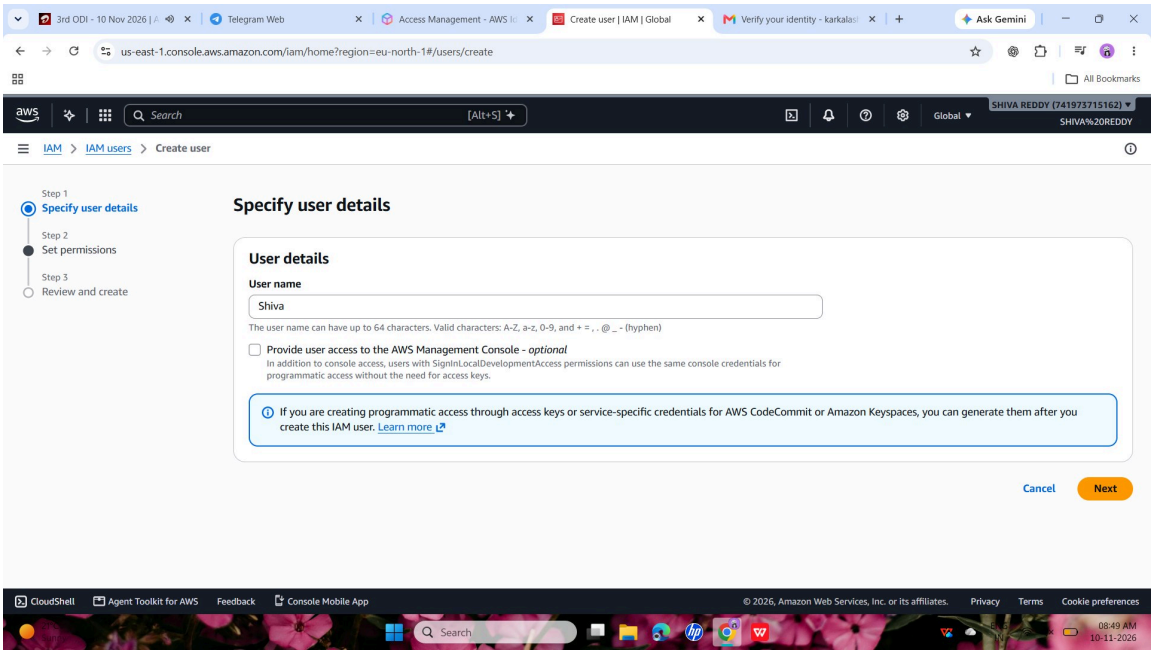


Fig 2: Create user - Step 1: Specify user details

In the first step of creating the user, the user name was entered as Shiva. The checkbox Provide user access to the AWS Management Console was kept unchecked so that the user is created with programmatic access only and no console password is generated.

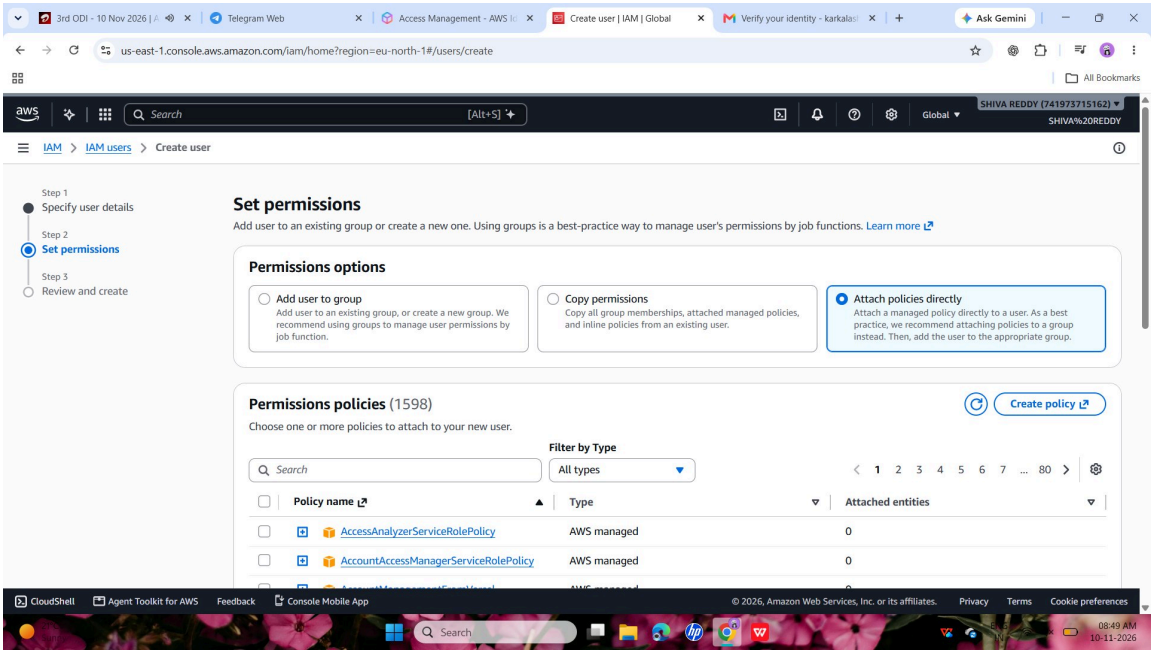


Fig 3: Create user - Step 2: Set permissions

In the second step, the option Attach policies directly was selected. The permissions policies list was searched and the AdministratorAccess AWS managed policy was selected to attach it directly to the user.

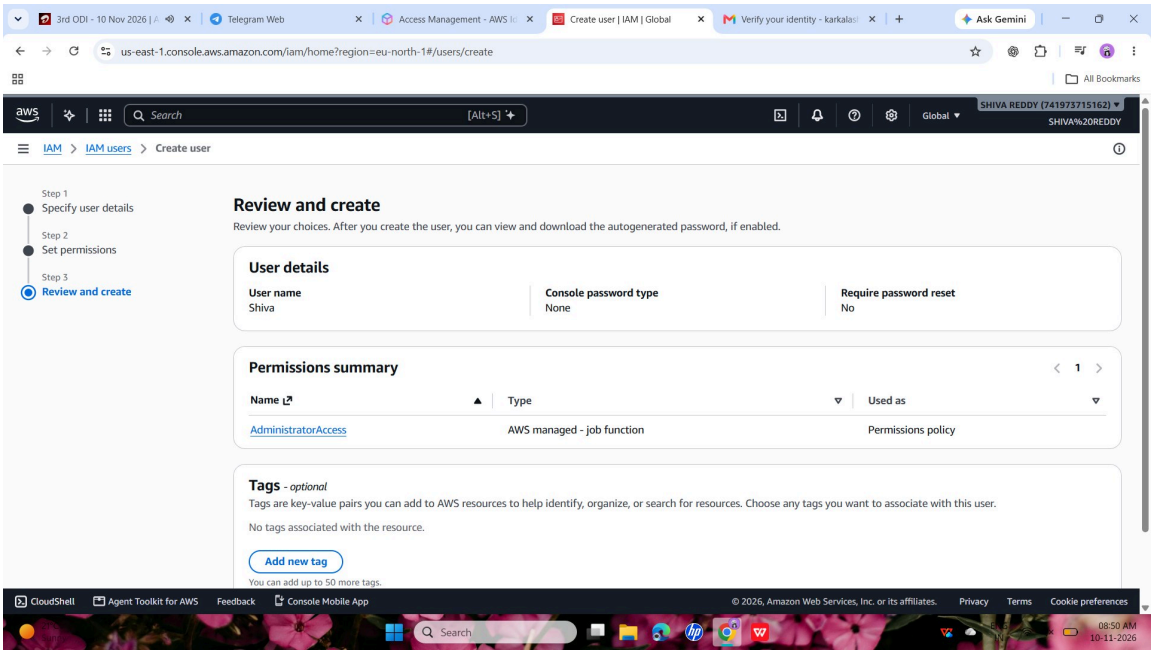


Fig 4: Create user - Step 3: Review and create

In the review step, the user details were verified. The console password type is shown as None and the AdministratorAccess policy is attached, which confirms that the user will have programmatic access only. The user was created after clicking Create user.

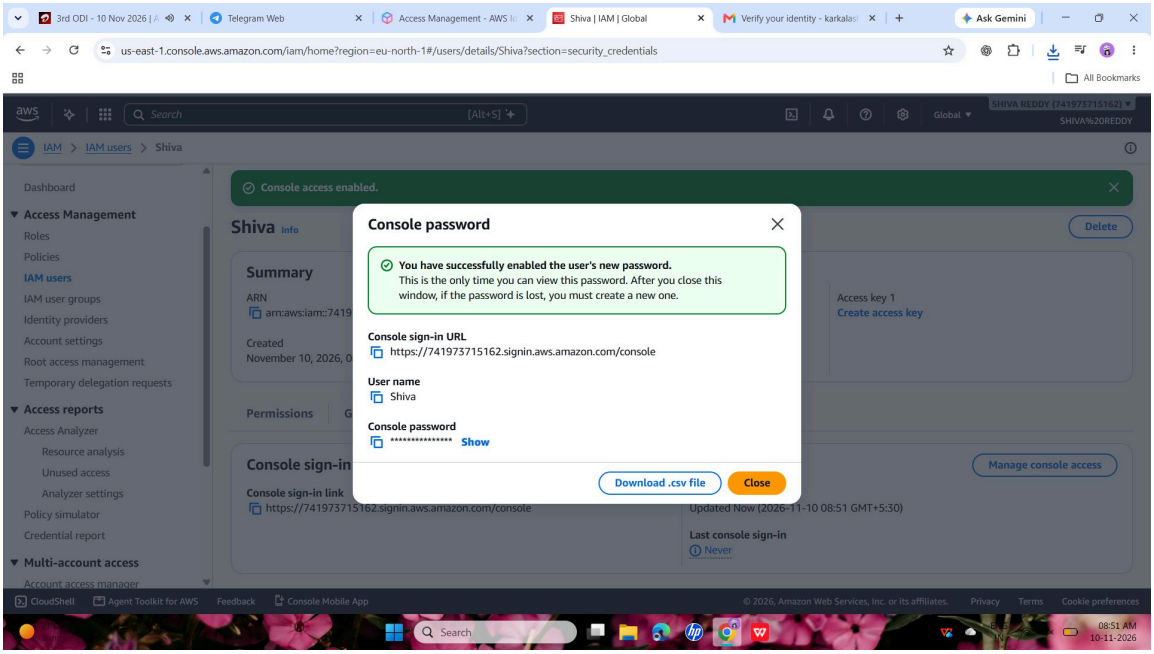


Fig 5: Console password window of the IAM user

To check that the user was created correctly, console access was enabled once from the Security credentials tab. The console password window displayed the sign-in URL along with the autogenerated password, and the credentials were downloaded as a CSV file. This was only for testing the user.

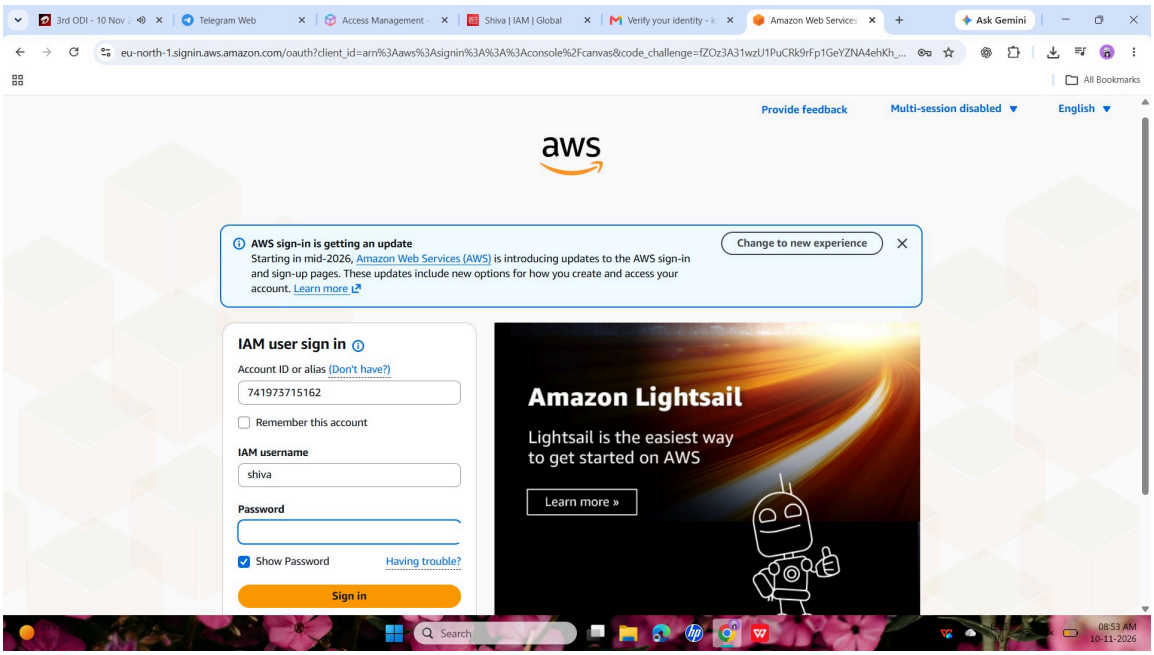


Fig 6: IAM user sign-in page

The AWS sign-in page of the IAM user. The account ID 741973715162 and the IAM user name were entered here to sign in as the IAM user and confirm that the user works.

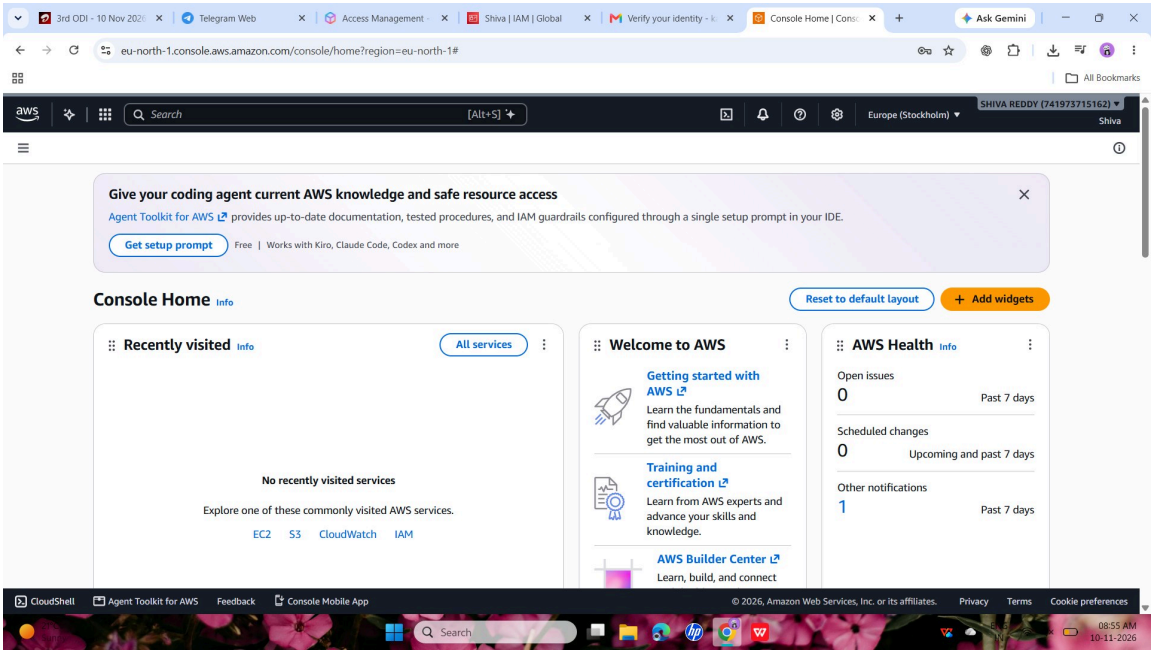


Fig 7: Console Home page after signing in as the IAM user

After signing in as the IAM user, the Console Home page opened successfully. The account is now accessed through the IAM user instead of the root account, which is a recommended security practice.

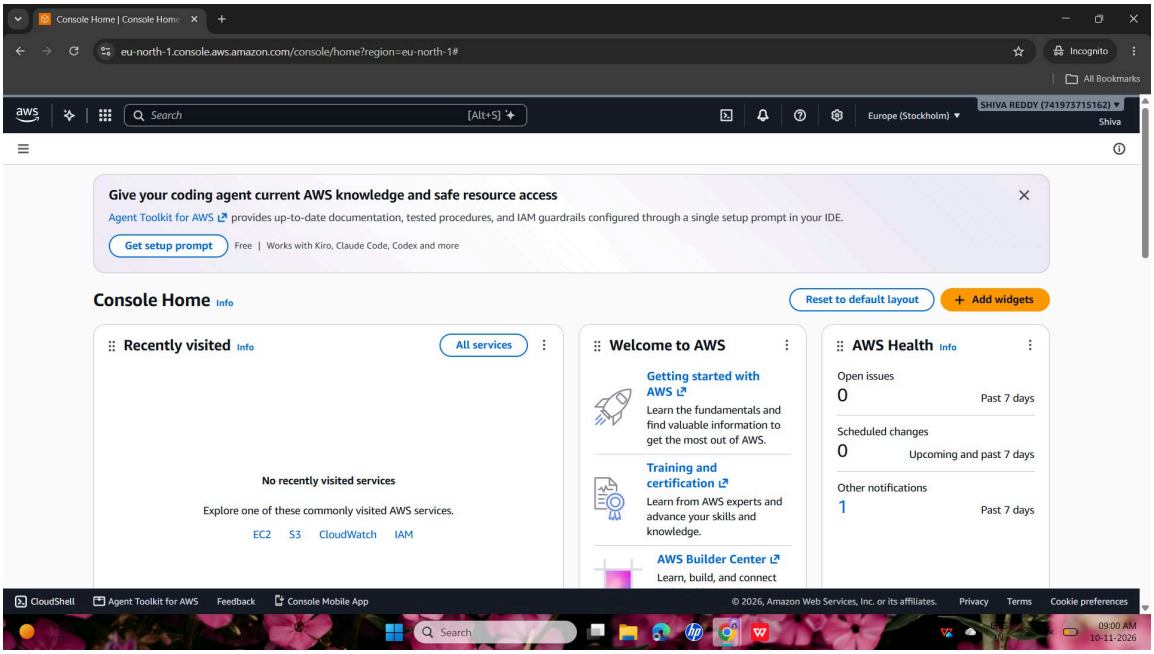


Fig 8: Console Home page of the signed-in IAM user

The Console Home page of the signed-in IAM user showing the recently visited services and the account details in the top right corner. The user is able to access the console normally.

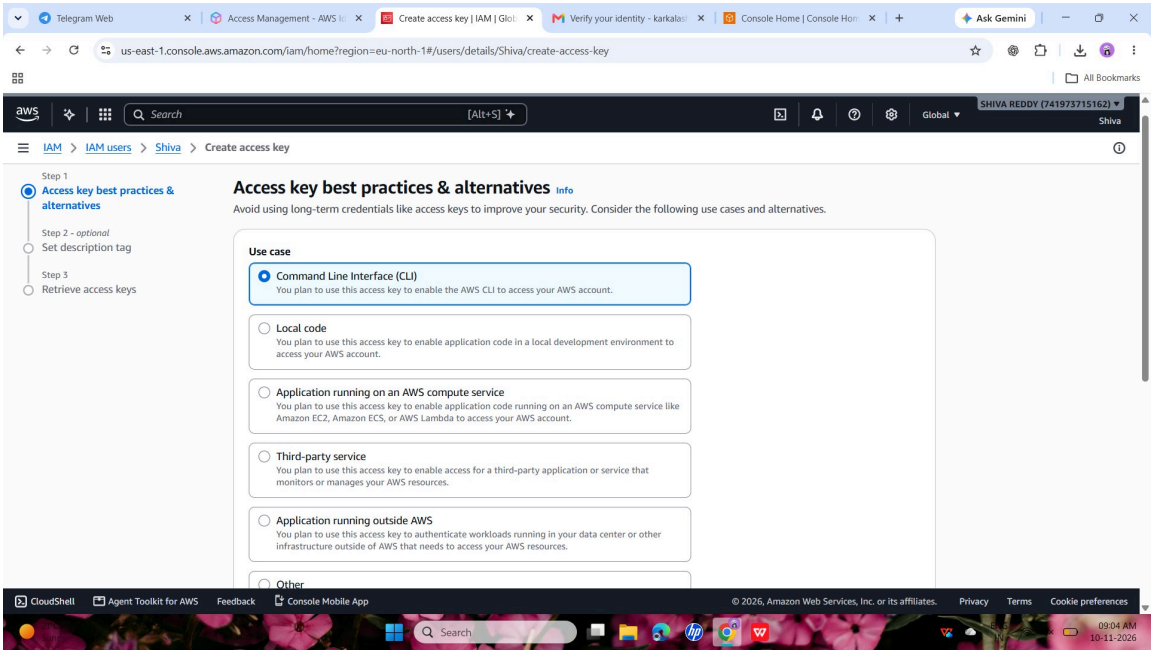


Fig 9: Create access key - Step 1: Selecting the use case

The Create access key page for the user Shiva. The Command Line Interface (CLI) use case was selected here, because the access key is used for programmatic access through the AWS CLI.

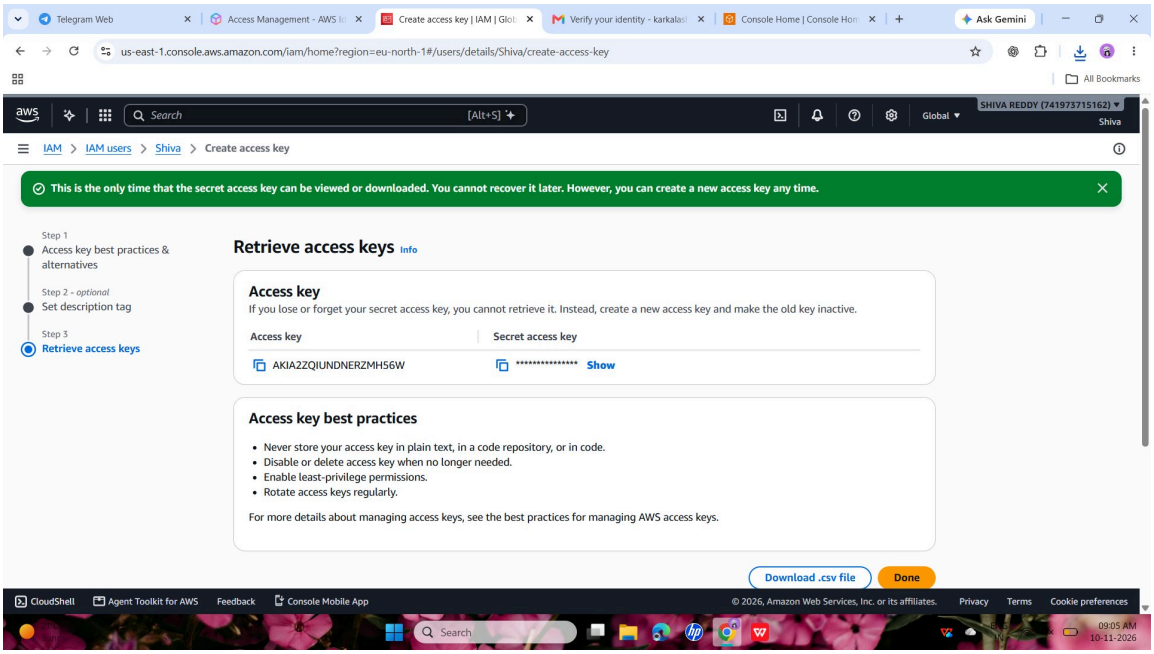


Fig 10: Create access key - Step 3: Retrieve access keys

The Retrieve access keys page displayed the Access Key ID and the Secret Access Key. Since the secret access key can be viewed only this one time, it was downloaded as a CSV file using the Download .csv file button.

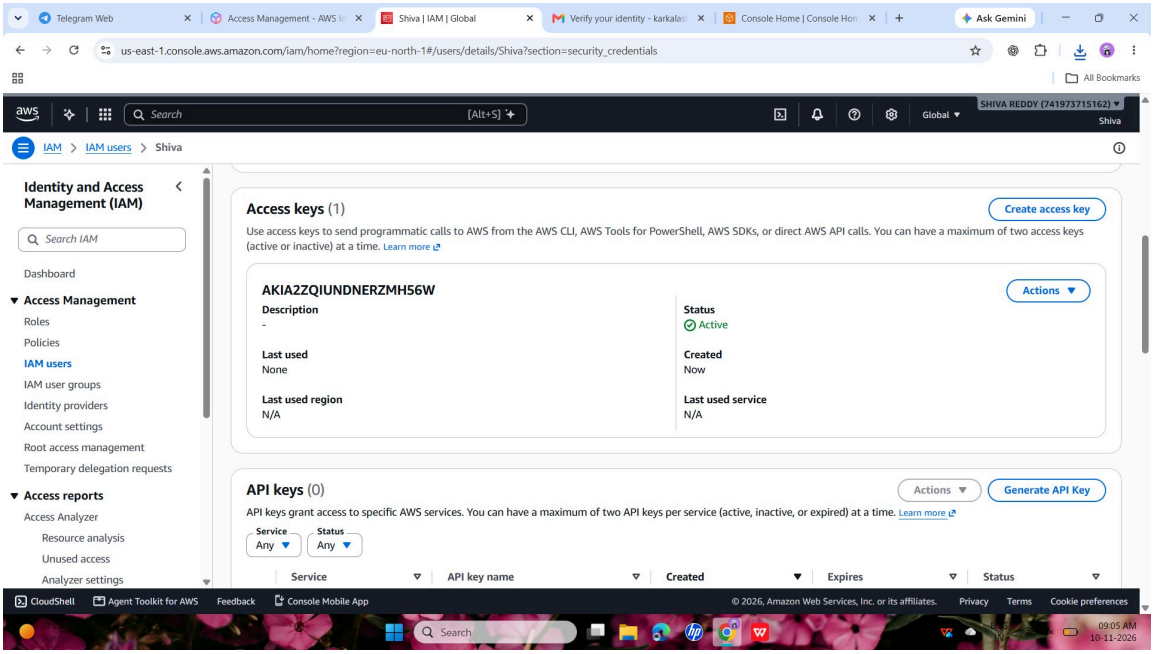


Fig 11: Security credentials page of the user Shiva

The Security credentials page of the user Shiva showing Access keys (1). The status of the key is Active and it was created just now, which confirms that the programmatic access for the user is ready.

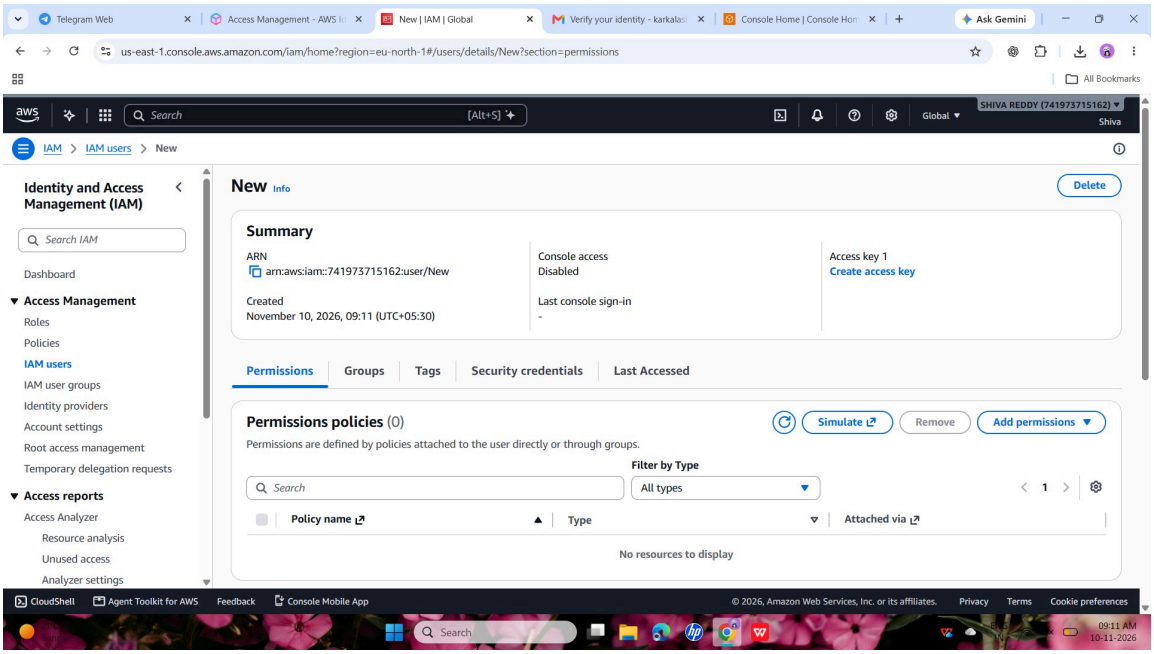


Fig 12: IAM user New details page

A second IAM user named New was created in the same way. The user details page shows that the console access is Disabled, which means this user has programmatic access only as required by the experiment.

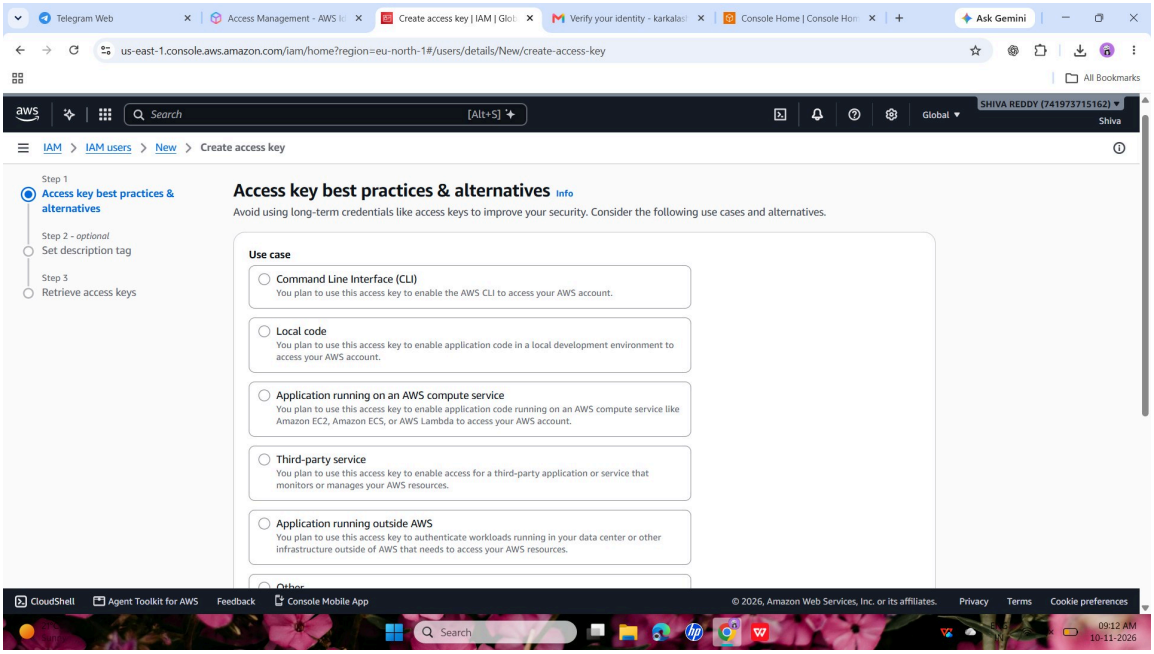


Fig 13: Create access key for user New - Step 1

The Create access key page for the user New where the use case is selected for generating the access key, in the same way as it was done for the first user.

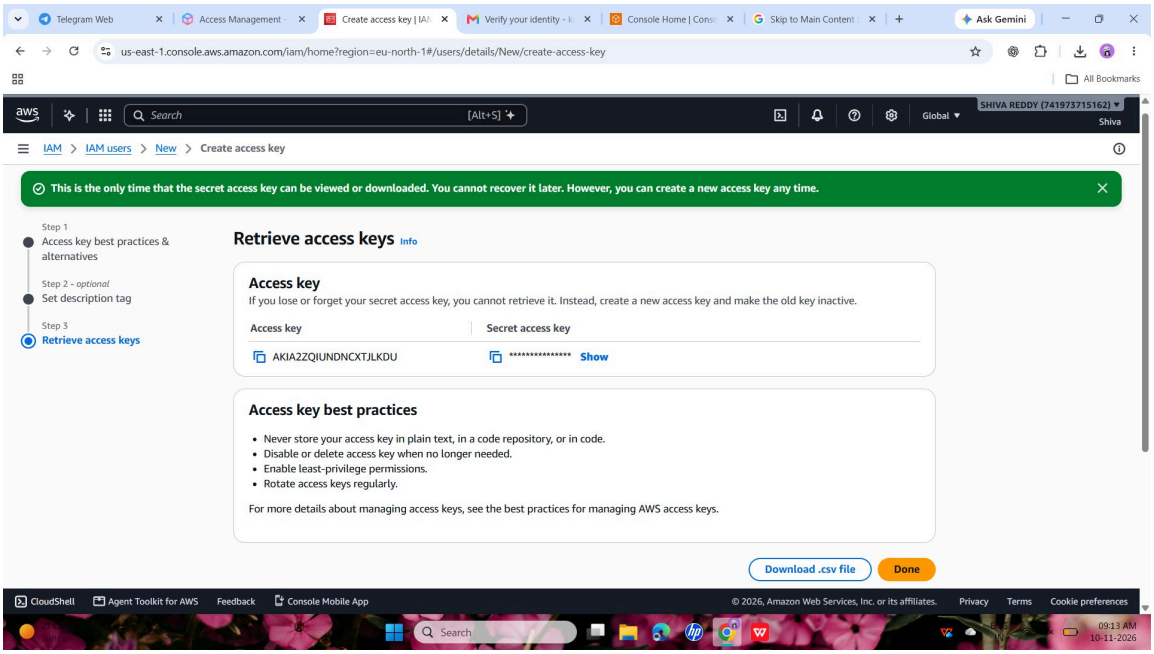


Fig 14: Retrieve access keys for user New

The access key of the user New was retrieved. The Access Key ID and the Secret Access Key were displayed only once and were downloaded as a CSV file for programmatic access.

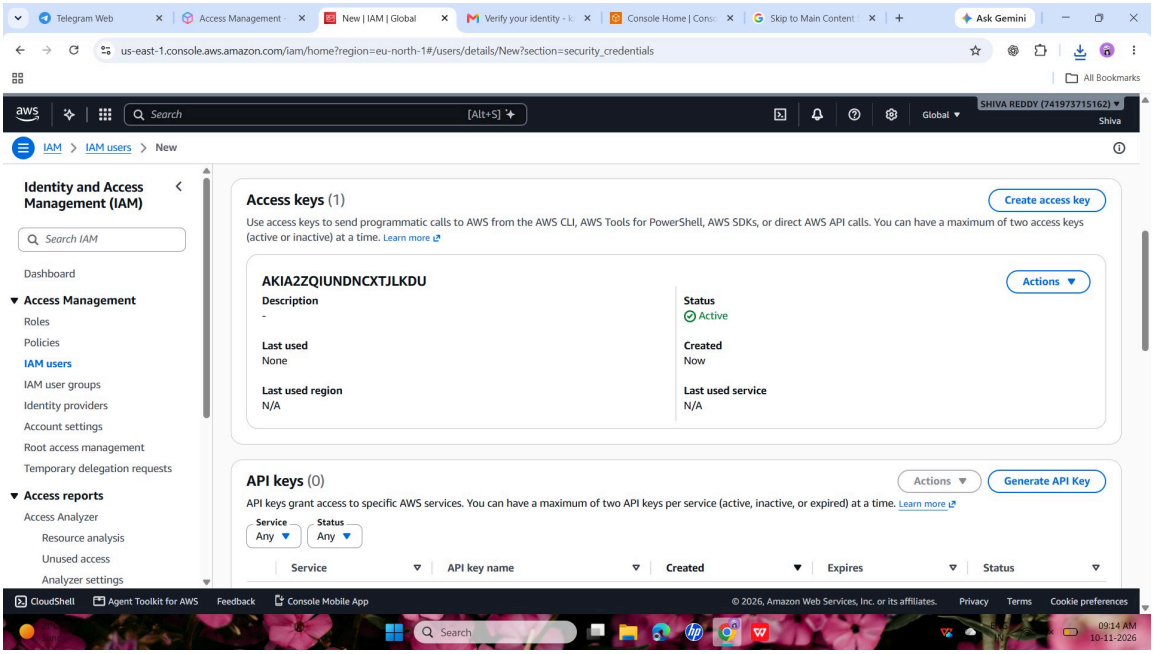


Fig 15: Security credentials page of user New

The Security credentials page of the user New confirming that the access key was created successfully with the status Active.

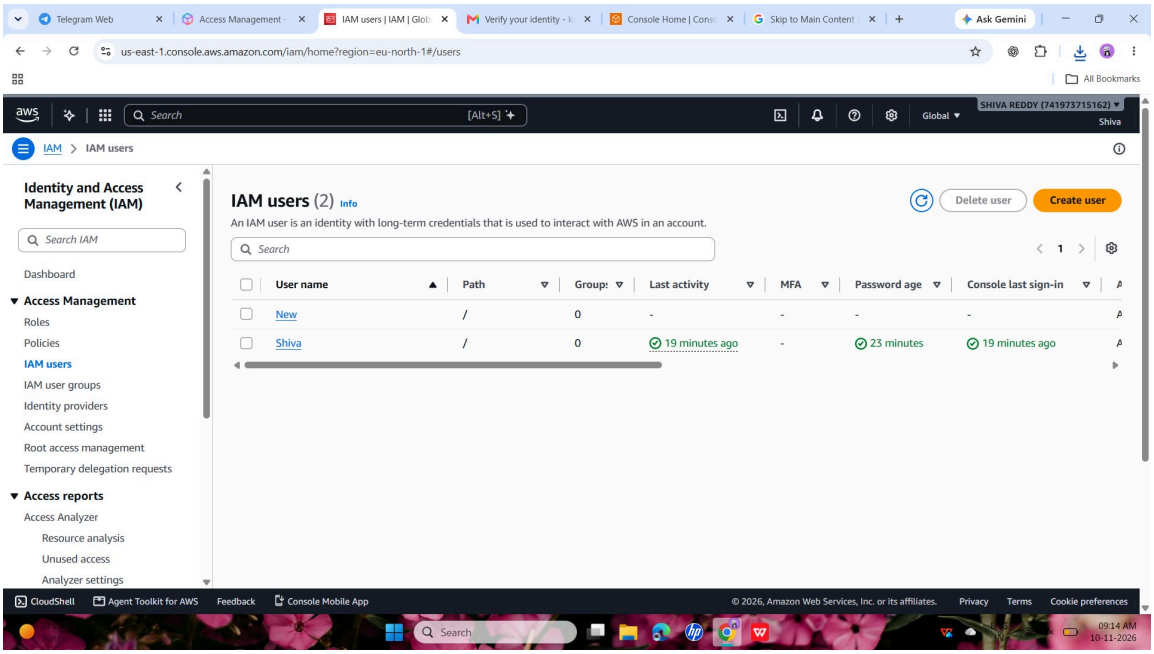


Fig 16: IAM users page listing both users

The IAM users page shows both the users New and Shiva created during this skill, along with their last activity and password age details.

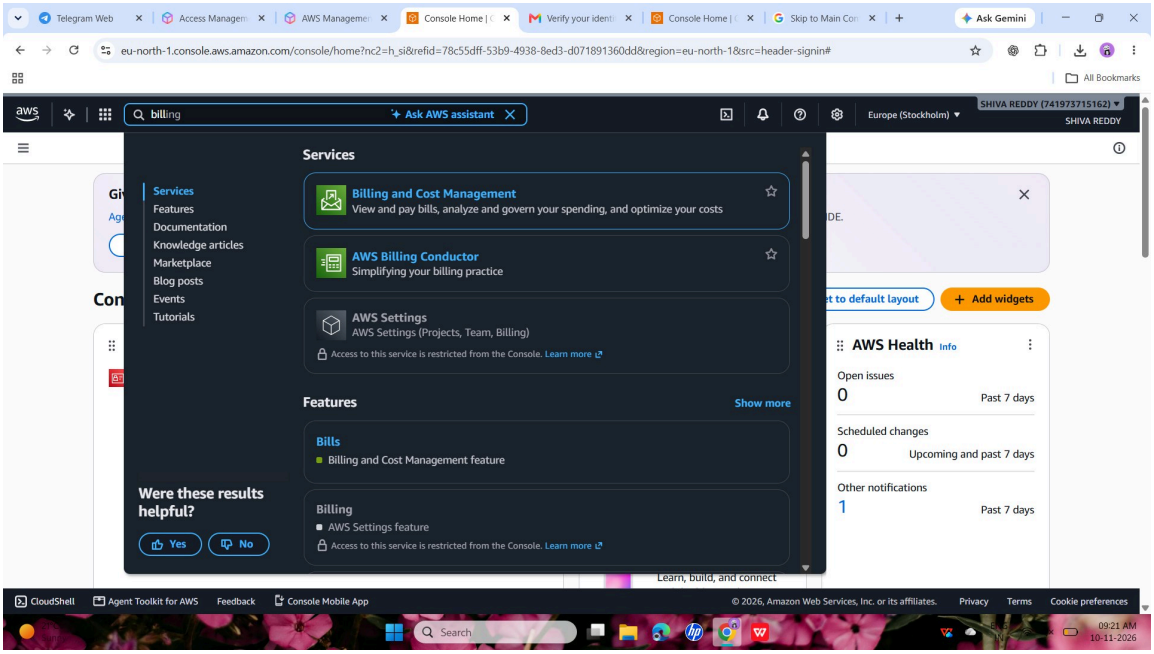


Fig 17: Searching for Billing in the AWS console

To set up the billing alert, billing was searched in the AWS console search bar and the Billing and Cost Management service was opened from the results.

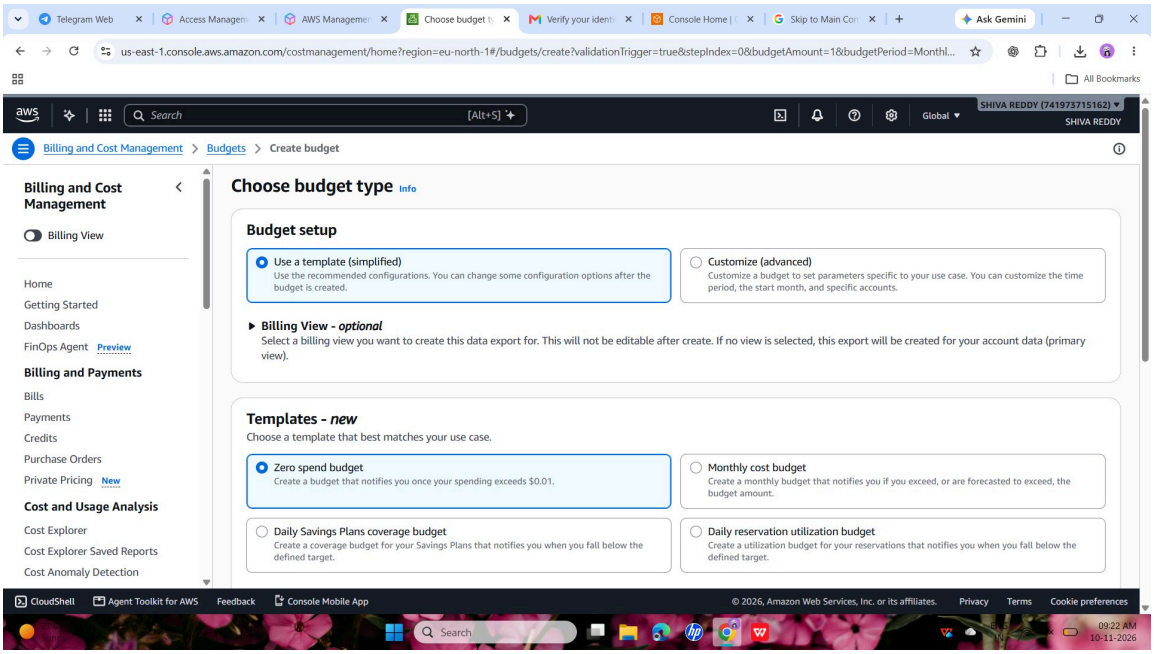


Fig 18: Create budget - Use a template (simplified)

In the Create budget page, the Use a template (simplified) option was selected first. The Zero spend budget template was chosen because it notifies by email once the spending exceeds $0.01, which is suitable for a free tier account.

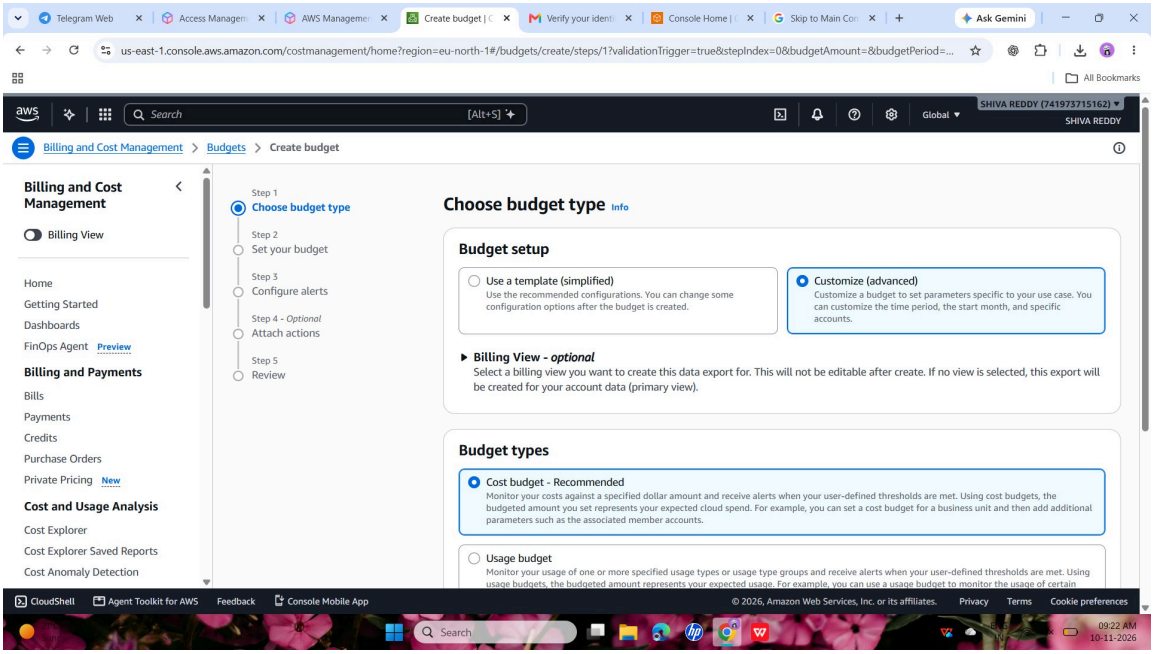


Fig 19: Create budget - Customize (advanced)

The Customize (advanced) option was also reviewed, where a Cost budget can be configured with a specific amount and time period. Finally the template option was used for simplicity.

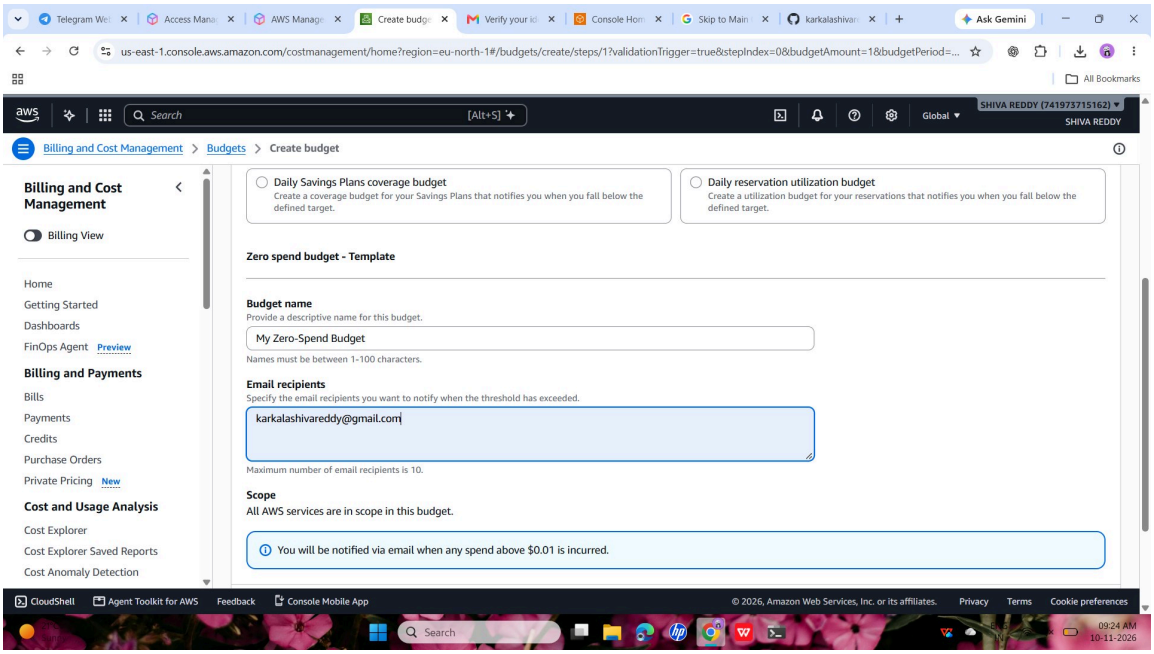


Fig 20: Zero spend budget configuration

The zero spend budget configuration. The budget name was entered as My Zero-Spend Budget and the email recipient was given as karkalashivareddy@gmail.com. All AWS services are in scope, so an email notification is received whenever any spend above $0.01 is incurred, which alerts before the free tier limit is crossed.